Biological Applications of X-ray Fluorescence Microscopy
Friday, August 9th 2013

Introduction to the Workshop				
8:40	9:00	Gayle Woloschak	Northwestern University	Introduction: Exploring "normal", "sick" and "artificial" in biological systems
Normal Cell Homeostasis				
9:00	9:30	Claire Weekley	University of Adelaide	Combining XFM and XAS to investigate the metabolism of selenium in biological systems
9:30	10:00	Robert Colvin	Ohio University	Using X-ray fluorescence microscopy to explore the unique subcellular distribution of Fe in primary cultured rat neurons
10:00	10:30	Carol Fierke	University of Michigan	Disruption of transition metal homeostasis by Cd: Implications for aging
10:30	10:45	**Coffee break/Poster viewing**		
Techniques I				
10:45	11:15	Patrick La Riviere	University of Chicago	3D X-ray fluorescence tomography using imaging detectors
11:15	11:45	Charlotte Gleber	Argonne Nat'l Lab	Developments in X-ray Fluorescence Tomography
11:45	12:15	Carol Hirschmugl	University of Wisconsin	Instrumentation Developments for Infrared Microscopy
12:15	1:30	**Lunch/Poster viewing**		
Disturbances in Homeostasis				
1:30	2:00	Martina Ralle	OHSU	Copper imaging in brains
2:00	2:30	Jane Flinn	George Mason University	Zn in behavior and macular degeneration
2:30	3:00	Tom O'Halloran	Northwestern University	Biomedical applications and metal homeostasis
3:00	3:15	**Coffee break/Poster viewing**		
Techniques II				
3:15	3:45	Richard Leapman	NIH	Nanoscale analysis of biological structures in the electron microscope using elastic and inelastic scattering
3:45	4:15	Lisa Miller	Brookhaven Nat'l Lab	
4:15	4:45	Brian Stephenson	Argonne Nat'l Lab	APS and APS upgrade
Poster Session Talks I				
4:45	5:00	tbd		Graduate student presentation
5:00	5:15	tbd		Graduate student presentation
5:15	5:30	tbd		Graduate student presentation
5:30	5:45	tbd		Graduate student presentation

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Techniques III and more

9:00	9:20	Si Chen	Argonne Nat'l Lab	The Bionanoprobe
9:20	9:40	Siwei Wang	Argonne Nat'l Lab	Data analysis for x-ray fluorescence microscopy: towards more automation
9:40	10:00	tbd		Postdoc presentation
10:00	10:20	tbd		Postdoc presentation
10:20	10:40	tbd		Postdoc presentation
10:40	10:55	**Coffee break/Poster viewing**		

Cancer Research, or Introducing the artificial materials into cells

10:55	11:25	Dev Mustafi	University of Chicago	Developing Targeted Vanadium-based Contrast Agents for Early Detection of Murine Colon Cancer Using Magnetic Resonance Imaging, X-ray Fluorescence Microscopy and a Novel Method of Image Co-registrations
11:25	11:45	Ye Yuan	Northwestern University	Targeting nanoparticles using cellular receptor EGFR
11:45	12:15	Lydia Finney	Argonne Nat'l Lab	Development of X-ray Fluorescence Capabilities at 8-BM
12:15	1:15	**Lunch/Poster viewing**		

Homeostasis Away from Mammalian Systems

1:15	1:45	Tracy Punshon	Dartmouth College	Using SXRF imaging in gene characterization
1:45	2:15	Matthew Marshall	Pacific Northwest Nat'l Lab	Toward High-Resolution Chemical Imaging of Biofilms: Integration of Synchrotron Imaging and Electron Microscopy
2:15	2:45	Ellery Ingall	Georgia Institute of Technology	Role of biogenic silica in the removal of iron from Antarctic Seas
2:45	3:15	Ben Twining	Bigelow Laboratory for Ocean Sciences	Studying the ocean through XRF analyses of its smallest residents
3:15	3:30	**Coffee break/Poster viewing**		

Poster Session Talks II

3:30	3:45	tbd		Graduate student presentation
3:45	4:00	tbd		Graduate student presentation
4:00	4:15	tbd		Graduate student presentation
4:15	6:00	Janos Kirz and James Penner-Hahn	Advanced Light Source; University of Michigan	Moderated Discussion and Workshop Conclusions